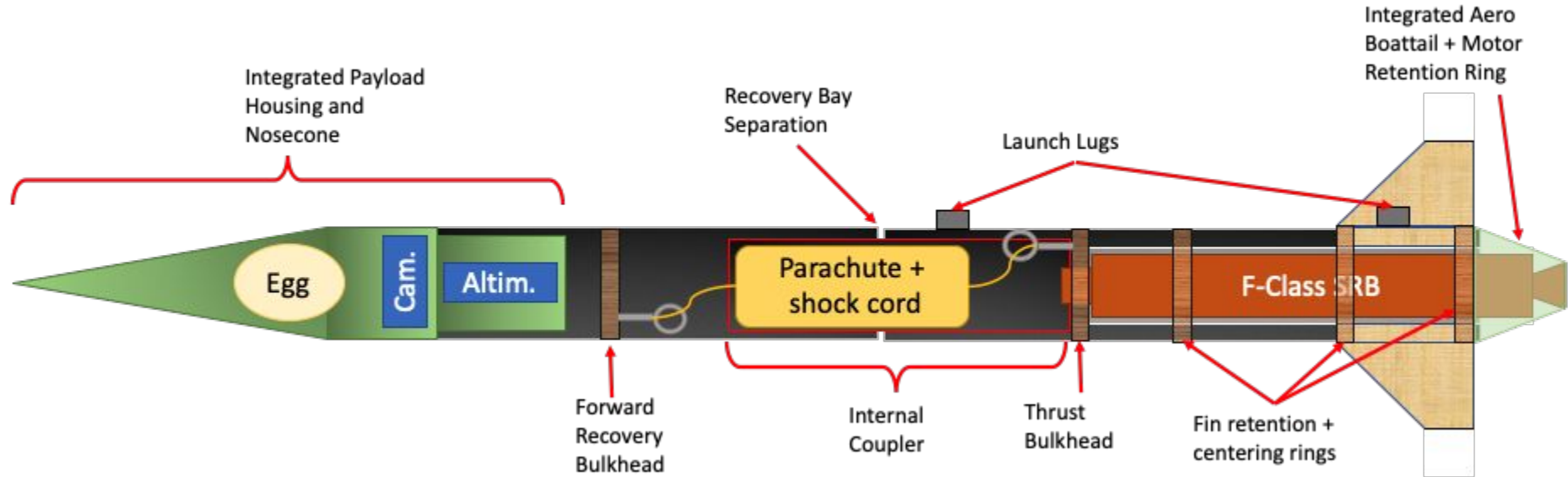


Vehicle Layout



Concept of Operations

Integration

1. Tie parachute into recovery bay, attach upper body tube to motor tube
2. Integrate motor and boat tail
3. (T-45 min to launch) Integrate payload into nose cone housing

Arming and Launch

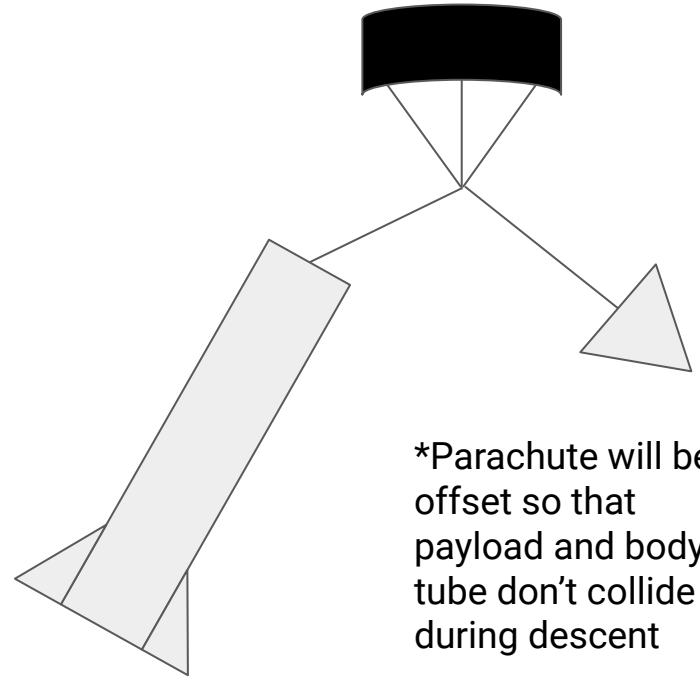
1. (T-20 min to launch) Arm camera and altimeter
2. (T-5 min) Insert igniter, rocket on rail, clear pad
3. Launch

Flight and Recovery

1. (T-0) Launch
2. (T+ ~2 sec) Motor Burnout
3. (T+ 8-10 sec) Apogee
4. (T+9-11 sec) Recovery Ejection
5. (T+ 1 min) Touchdown

Recovery Plan

- Ejection due to black powder charge in motor, occurs 6 seconds after burnout
 - Velocity at deployment = -0.5 [ft/s]
- Current plan is for 1 main parachute
 - Will assume $C_D = 1.2$
 - $D = 18''$
- Will size for a ~20-30 ft/s descent velocity
 - Current expected descent velocity = 22.4 [ft/s]

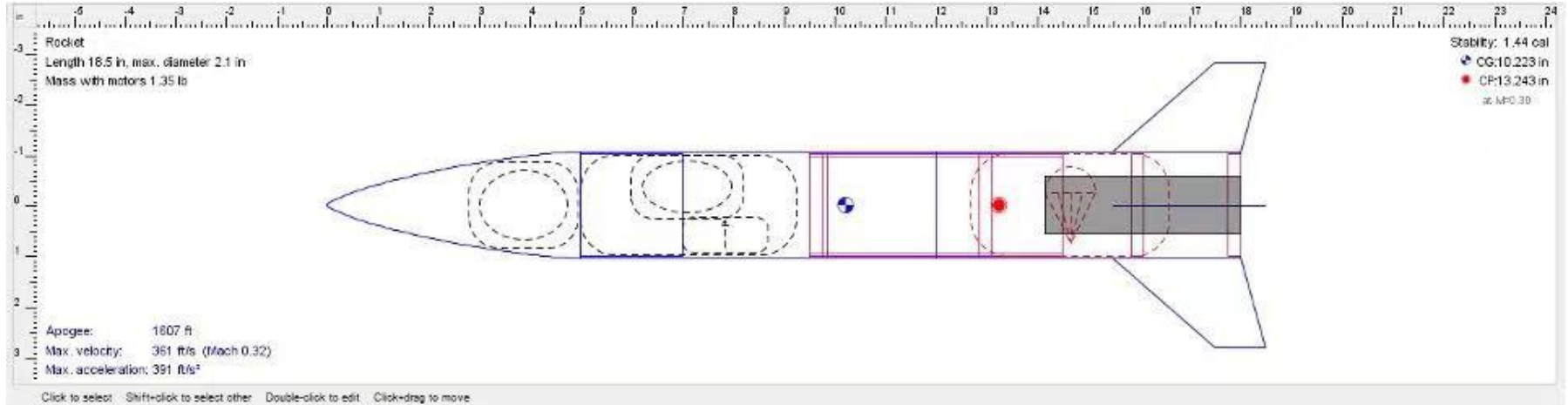


Stability Analysis

By using Barrowman's equations, here we can set up the calculations for the total coefficient of normal force (C_{Naf}) and the locations of C_p and C_g , and static margin

```
1 clear all; clc; close all;
2
3 % initial conditions for trapezoidal fin set, in inches
4 Ln = 5;
5 D = 2.1;
6 Cr = 2.5;
7 Ct = 1;
8 S = 1.75;
9 N = 4;
10 Xs = 2;
11
12 C_Nan = 2; % elliptical nose cone
13 alpha = 0.333;
14 beta = 16; % for N = 4
15 X_N = alpha * Ln;
16
17 Kfb = 1 + (D/(2*S+D));
18 Lf = (S^2+(Xs+Ct/2-Cr/2)^2)^(0.5);
19
20 C_Naf = (beta*(S/D)^2)/(1+(1+((2*Lf)/(Cr+Ct))^2)^0.5); % Normal force
21
22 C_Nafb = Kfb*C_Naf;
23
24 XF = 15.5;
25 xf = XF + (Xs*(Cr+2*Ct))/(3*(Cr+Ct)) + (1/6)*(Cr+Ct-(Cr*Ct)/(Cr+Ct));
26
27 C_NaT = C_Nan + C_Nafb; %total normal force on rocket
28 X_cp = (C_Nan * X_N + C_Nafb * xf)/C_NaT; % cp of entire rocket
29 X_cg = 10.223;
30 Static_margin = (X_cp - X_cg)/D;
```


Stability Analysis



Elliptical nose cone with 4 fin sets at 0.3 Mach:

CP	10.22
CG	13.24
Static Margin	1.438
C_Naf	4.30
C_Nat	7.91

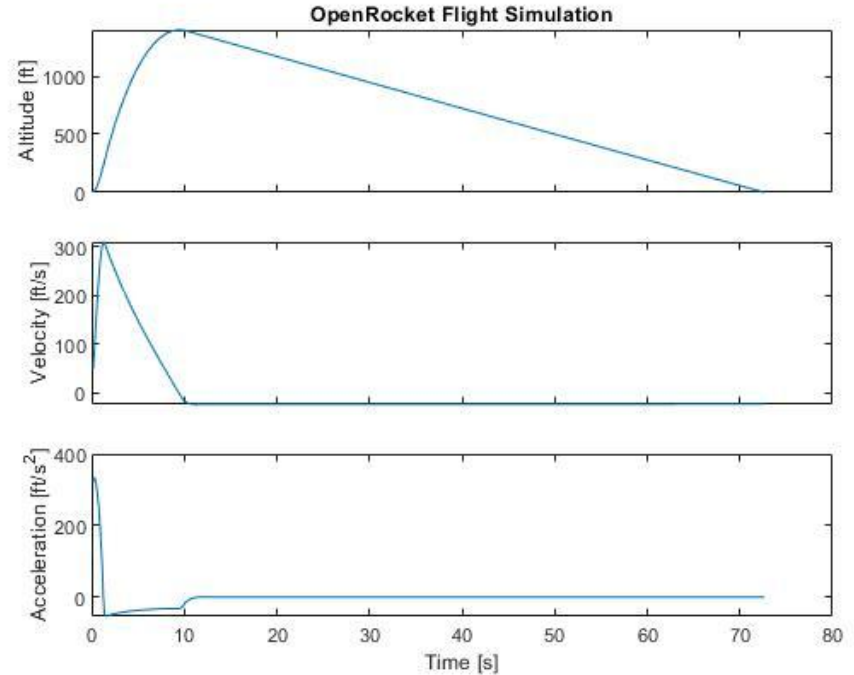
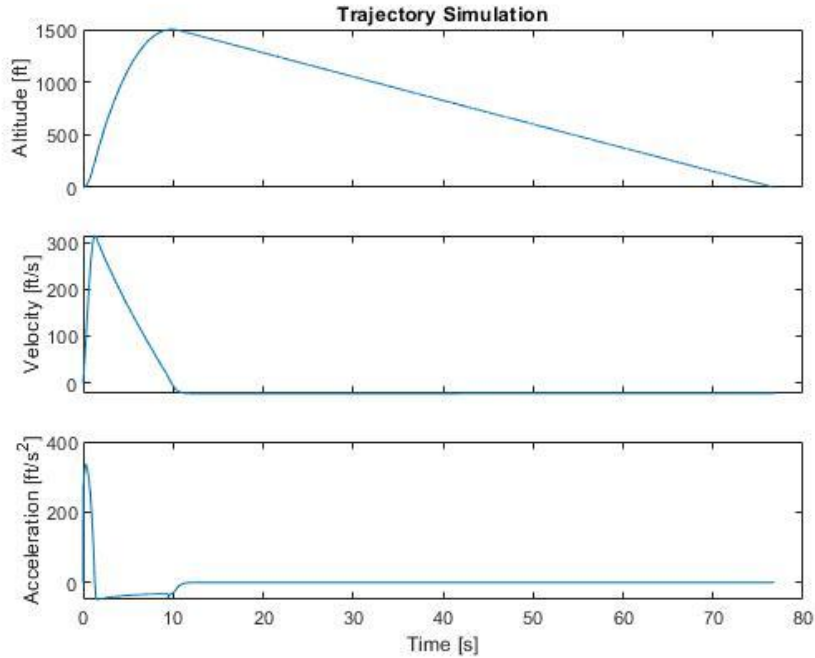
Therefore as long as we meet our project specs, this should be stable

Trajectory Analysis

Flight Specification	Trajectory Code Results	OpenRocket Simulation Results
Apogee (ft)	1503.6	1400.2
Time to Apogee (sec)	9.8	9.5
Time to Burnout (sec)	1.3	1.3
Burnout Altitude (ft)	244.4	234.3
Burnout Velocity (ft/sec)	314.3	310.3
Descent Velocity (ft/sec)	-22.4	-22.1

* 180g mass was added at CG location of OpenRocket model to give it the same dry mass as the MEL for simulation

Trajectory Analysis



Trajectory Analysis

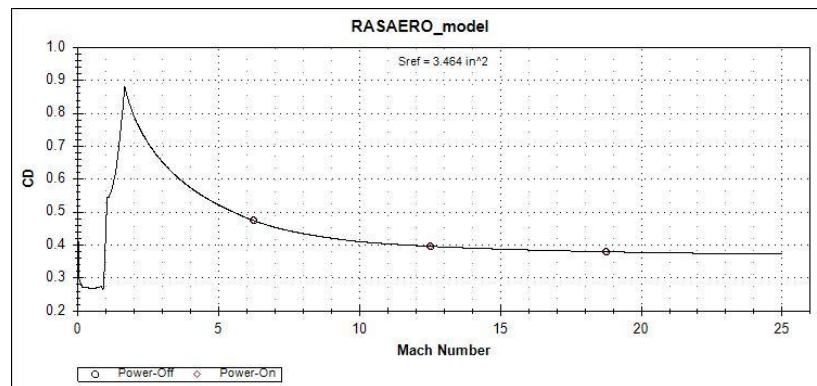
The rocket model was recreated in RASAero, which gave drag coefficients as a function of Mach number.

- Will use drag results from testing in final version.
- Atmos.mat file provided for Lab 2 used for atmospheric variation.

Currently not considered in the trajectory:

- 3-D effects
- Wind
- Lift

C_d Plot



Preliminary Structural Analysis: Loads and Allowables

Loads		
Loadcase	Load (lbf)	Load (G's)
Ignition/Max Thrust	18.00	12.17
maxQ	2.02	10.69
Burnout	2.02	-2.48
Recovery Deployment	11.17	8.17

Joint and Interface Load Allowables

Joint/Interface	Thrust bulkhead/body tube	lower body tube/coupler	upper body tube/recovery bulkhead	Payload holder/body tube	payload holder/nose cone
characteristic area (in^2)	0.79	12.57	0.79	0.05	0.05
stress allowable (psi)	1000.00	1000.00	1000.00	25000.00	25000.00
Load allowable (lbf)	785.38	12566.00	785.38	1300.00	1300.00
Supported mass (lbm)	1.37	1.13	0.79	0.68	0.24

Preliminary Factors of Safety

- Based on preliminary analysis, all factor of safety are well in excess of typical flight values (FoS ~ 2), indicating all structures will survive expected flight and recovery loads
- Will not pursue efforts to reduce unnecessary structural mass, since sizing is primarily based on available materials and handling loads

Preliminary Factors of Safety for Flight Load Cases at All Major Vehicle Joints and Interfaces

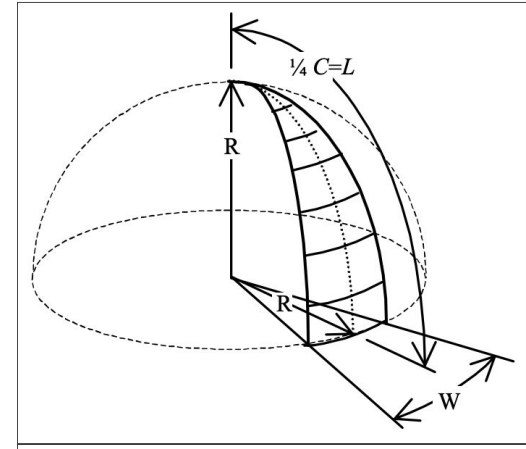
Supported mass (lbm)	Thrust bulkhead/body tube	lower body tube/coupler	upper body tube/recovery bulkhead	Payload holder/body tube	payload holder/nose cone
Ignition/Max Thrust	44	915	82	157	441
maxQ	54	1042	93	178	502
Burnout	232	4492	402	768	2167
Recovery Deployment	70	1363	122	233	657

Manufacturing Plan - Rocket Body & Aerostructures

- Tube
 - CF, double layer
 - 0.046 wall thickness
 - 11" + contingency
 - ~ day of work
- Coupler
 - CF, double layer
 - 4" + contingency
- Fins/Bulkheads
 - Plywood
 - Laser cut
 - .dxf files
- Nosecone and payload housing
 - 3D printed
 - Nosecone submitted & printed
 - Housing to be printed
- Next steps
 - Analyze loading on bulkheads
 - Save weight w/balsa wood
 - Analyze loading on coupler
 - Save weight w/single layer

Manufacturing Plan - Recovery System

- Parachute - hemispherical
 - 18" diameter
 - 22.4 ft/s descent velocity
 - $CD = 1.4$
 - 8 gores
 - 7 in width, 14 in length
 - 2" diameter spill hole
 - Nylon
 - Source: Amazon
 - Ottertext 1.9 Oz Ripstop Nylon
 - \$5.45/yd
 - Sold by the yard, 60" wide
- Shock Cord
 - 2.25 ft (~1.5x parachute diameter + contingency)
 - Source: Amazon
 - BENGKU Outdoor Mil-SPEC 550lb Paracord
 - \$9.89/100 ft



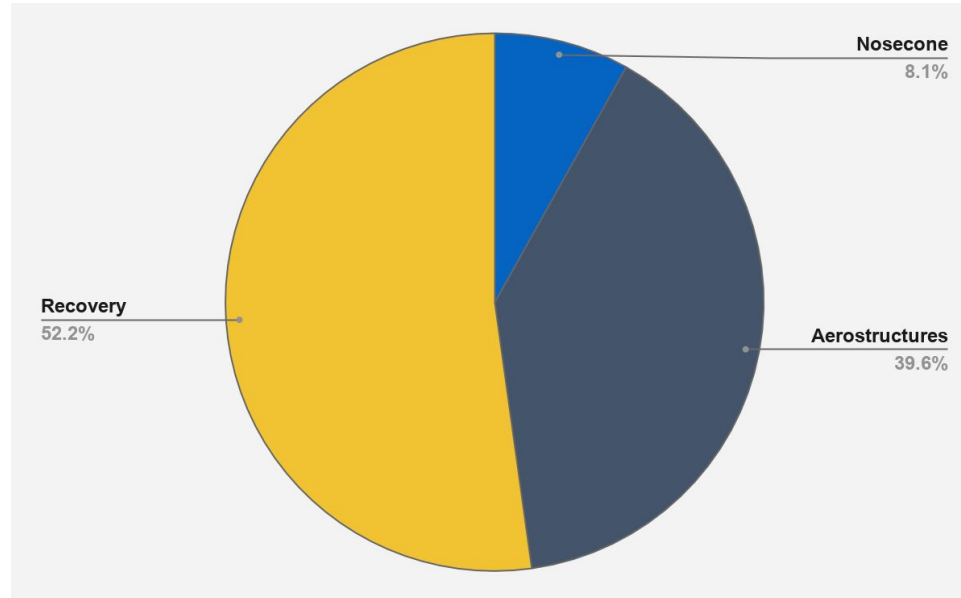
- Next Steps
 - Analyze semi-hemispherical parachute
 - Improved descent velocity-weight ratio

Preliminary Budget

Component	Size	Unit	Required Materials	Quantity	Price
Nosecone	3.24	hr	PLA	1	\$1.62
Payload Holder	3.24	hr	PLA	1	\$1.62
Body Tube	80.58	in2	Total	--	\$11.03
			Mandrel (in)	12	\$1.20
			Release film	2	\$1.61
			Prepreg	1	\$7.25
			Peel ply	1	\$0.48
			Breather	1	\$0.24
			Vacuum bag	1	\$0.24
Forward Recovery Bulkhead	3.93	in2	Plywood	1	\$0.08
Parachute	0.75	yds	Nylon	2	\$10.90
Shockcord	6.33	ft	Parachord	1	\$9.89
Coupler	29.30	in2	Total	--	\$4.08
			Mandrel (in)	5	\$0.50
			Release film	2	\$0.59
			Prepreg	1	\$2.64
			Peel ply	1	\$0.18
			Breather	1	\$0.09
			Vacuum bag	1	\$0.09
Thrust Bulkhead	3.93	in2	Plywood	1	\$0.08
Forward Centering Ring	3.93	in2	Plywood	1	\$0.08
Lower Centering Ring	3.93	in2	Plywood	1	\$0.08
Fins	4.56	in2	Plywood	4	\$0.36
Total					\$39.82

	Expected	Budgeted	Margin	% Margin
Cost	\$39.82	\$50.00	\$10.18	25.58%
Print Time (hrs)	6.48	10	3.52	54.32%

Budget Breakdown



Subsection	Cost	Print Time	Contribution to Total Cost	Contribution to Total Print Time
Nosecone	\$3.24	6.48	8.14%	100.00%
Aerostructures	\$15.79	0	39.65%	0.00%
Recovery	\$20.79	0	52.22%	0.00%

Testing Plans

- Parachute Descent Velocity
 - Compare theoretical to actual (desired <30 ft/s)
 - Test our manufactured parachute
 - Test if payload survives
- Wind Tunnel Testing
 - Stability Test
 - Clamp rocket at C_G
 - Find experimental C_P
 - Drag Test
 - Measure drag force using the load cell
 - Find experimental C_D

Testing Plans (cont.)

- Structural loading - Test structure under highest loads
 - Recovery load testing
 - Tension test of shock cord
 - Predict tension on shock cord based on predicted deceleration
 - Thrust load
 - Compress body tube using an equivalent weight
- Weighing the rocket
 - For trajectory analysis
 - Experimentally find C_G with a balance test



